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Seminar in Empirical Finance and Financial Econometrics

README on the Seminar Project

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March 7, 2025

Seminar Project

Preliminaries

- ▶ This assignment can be done either *individually*, or in a *group of 2 or 3 students*
- ▶ Select a research question
 - ▶ Should be related to *financial econometrics*, *empirical finance*
 - ▶ A good starting point could be to take any methodology considered in class, or take a paper from the reading list, and build a sensible empirical analysis around it
 - ▶ Selected research question must be approved in advance: approach me after the class (one feedback loop: thesis proposal submission on 05/01)

Seminar Project

Preliminaries

- ▶ Three “check points”:
 1. Approval of the research question for the project (mandatory, but is not graded), approach me in class, make sure your thesis proposal will be approved
 2. A project presentation during the seminar: present your first results in class: everybody has to present a part of the project.
 3. Project paper

A Question for the Project

Basic requirements

- ▶ What is taken into account for approving a question
 - ▶ should **imply a certain level of complexity** for the empirical analysis (at the same time, the question itself might be very simply formulated, see examples below...)
 - ▶ the proposed analysis should be [somewhat] **motivated** (this should indicate that students understand what can be expected from the analysis)
 - ▶ ideally, the proposed question should be **supported with some basic literature background** (e.g., few relevant research papers)

A Question for the Project

Examples of valid project questions (I)

- ▶ When intra-daily information is useful for volatility forecasting?
 - ▶ Consider GARCH(1,1) and Realized GARCH(1,1) models (we discuss them during the seminar meetings). The former is based on the low frequency (usually, daily) returns. The latter also exploits realized measures of return variance that can be estimated with high frequency (intra-daily) data. The difference in model performance can be directly related to this fact.
 - ▶ Take a sample of several stocks and consider relatively long sample period (e.g., 10 or 15 years)
 - ▶ Estimate both models and compare (statistically) the quality of, say, one-day ahead variance forecasts for the selected assets. This comparison can be also done within subperiods (e.g., crisis times vs periods of mild or low volatility), where the difference in model performance might vary.
 - ▶ This might help to understand at what circumstances and by how much (if any) intra-daily information improve the volatility forecasts within the GARCH-type framework.

A Question for the Project

Examples of valid project questions (II)

- ▶ Exploring the limits of the Realized Variance (RV) estimator
 - ▶ The idea is to design a proper empirical analysis and understand the limitations of the RV estimator constructed from high frequency data (we discuss this estimator later in the seminar)
 - ▶ Take a sample of several stocks (preferably, with different average liquidity) and consider intra-daily prices for a sequence of trading days (say, take few months of data). Ideally, if several sub-periods are considered, e.g. when volatility level is generally high and low.
 - ▶ Construct RV estimates for the selected stocks and trading days. Construct RV at several different sampling frequencies (say, ranging from 1sec to 30min) and construct volatility signature plots for each considered stock and trading day.
 - ▶ Analyze the results and identify the range of frequencies which exhibit presumably consistent RV estimates. Discuss how this range depends on liquidity (comparison across stocks) and overall volatility level (comparison across trading days).

A Question for the Project

Examples of valid project questions (III)

Diffusion Index forecasts of stock returns

- ▶ Take a sample of several stocks, variables of interest that you aim to forecast
- ▶ Take a high-dimensional data set, e.g. macroeconomic data, or financial data and estimate factors, e.g. considering different methods for factor extraction, principal components, maximum likelihood etc. (we will learn about that in the seminar)
- ▶ Perform an out of sample forecasting horse-race
 1. Regressing on factors extracted by MLE,
 2. Regressing on factors extracted by PCA
 3. Regressing on all principal components, choose by LASSO

A Question for the Project

Examples of valid project questions (IV)

Compare Generalised Dynamic Factor Model and static factor models in a real time pseudo out of sample forecasting evaluation:

- ▶ Take a sample of several stocks, variables of interest that you aim to forecast
- ▶ Extract factors from a high-dimensional panel of time series, try different methods (we will learn about), extending with squares, selection of variables etc.
- ▶ Perform an out of sample forecasting horse-race
 1. Regressing on static factors (from cross-section)
 2. Regressing on static factors + lags
 3. Regressing on a state estimated for static factors
 4. Regressing on dynamic factors (extracted by dynamic principal components)

Project presentations

Expected structure

- ▶ Clearly **formulate a research question** (see examples above)
 - ▶ explain what you are about to do, main objectives of the empirical analysis
- ▶ Present a **roadmap** of the project
 - ▶ relevant literature (at least, few papers)
 - ▶ general strategy for the analysis (methodology, econometric strategy, statistical evaluation of results, robustness analysis, ...)
 - ▶ data that will be used (sources of data, what particular sample has been selected, etc.)
 - ▶ ...
- ▶ **Preliminary results** (if any preliminary analysis has been performed by the presentation date)
 - ▶ nice if results are presented graphically on your slides
 - ▶ discussion of the following steps
- ▶ Conclusion
 - ▶ explain intuitively what are the **expected findings** of the project work (some might be supported by the preliminary analysis)
 - ▶ try to discuss main expected takeaways of the analysis
 - ▶ ...

Project Paper (due 08/01)

General tips

- ▶ Brief paper (12-15 pages), but well structured
 - ▶ Structure should resemble a typical research paper (see many examples in the reading list)
 - ▶ ...and must not look like a technical report with dry equations and output results with a minimum of text
- ▶ Some important tips:
 - ▶ Paper should contain well explained (and well designed !) graphical illustrations / tables / estimation results, etc. (please, avoid print screening of your estimation output, construct clean tables and plots instead using graphical packages)
 - ▶ Figure “↗” Table. However, all tables / figures have to be carefully interpreted and explained in the main text.
 - ▶ Providing intuition is always useful. Please, make sure that you carefully discuss and interpreted your empirical findings in the project paper. You should demonstrate that you understand what you do and why.

Project Paper (due 08/01)

Expected structure (very roughly): section by section

► Introduction

- Clear description of the research question, main objectives
- Brief literature review
- Very brief discussion of methodology, data and empirical strategy (what you actually do in your analysis)
- Brief outline of main results

► Methodology and empirical strategy

- Detailed description of the methodological approach (discuss the choice and relevance of the selected methodology, describe the model or framework that is used for the analysis, describe your empirical strategy: explain what you are going to estimate / forecast / test / etc. and how you plan to perform it)

► Empirical results

- Describe the data that you will use in your analysis (sources, sample selection, etc.)
- Present results of the empirical analysis (depending on the question, it might be estimation results, forecasting results, results of the statistical model comparison, etc.). Use graphical visualisation of your results and properly designed tables – it should be easy to read and understand. Carefully discuss the findings and explain how these results respond to your research question.

► Outline

- Main results, conclusions and take-aways from the performed analysis.
- Discuss possible extensions, limitations (if any) of your analysis and open questions ...